Research Project: Using MCDA tools to solve a decision problem

Multiple Criteria Decision Support Systems

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12th June 2025

1 PART 1: Definition of Decision Problem

1.1 Definition of the decision maker

The decision maker (DM) in this case is a master's university student who is considering going on an exchange program to a university abroad. The DM's primary goal is to have a unique and enriching experience while making the most of the experience.

Given the wide range of universities and locations available, the DM faces a complex decision-making process with numerous options to choose from. This overwhelming number of options, combined with the significant impact the decision will have on the DM's academic and personal life, makes it crucial for the DM to evaluate each option thoroughly in order to make an informed choice.

As the DM is uncertain where to begin their research to decide where to make the exchange experience, the goal is to **identify the most suitable alternatives**, focusing on the most widely preferred exchange universities to help the DM explore the best options based on their values. With many factors to consider, like reputation, cost of living, language compatibility, and cultural opportunities, the DM isn't sure where to begin their research.

1.2 Criteria Definition and Performance Variables

When students choose an exchange university, they're looking for more than just a place to study—they want an experience that checks all the right boxes for their future. This section defines the key factors they prioritize: university reputation, financial support, living costs, and long-term considerations like quality of life and career prospects.

The criteria are:

- **Reputation of the University**: This variable measures the global reputation and academic standing of the university as its position to university rankings. The scale ranges from 1 to 500, where 1 represents the highest reputation. The goal is to **minimize** this criterion to ensure a prestigious and recognized academic experience.
- **Scholarships and Financial Support**: This criterion evaluates the financial support available for the exchange program, such as Erasmus+ scholarships or other funding options. The scale ranges from 0 to 15,000€, with higher values being more favorable. The goal is to **maximize** this criterion to reduce the financial burden on the student.
- **Monthly Cost of Living**: This variable represents the estimated monthly expenses (housing, food, transportation, etc.) while studying abroad. The scale ranges from 500€ to 2,500€, where lower costs are preferred. The goal is to **minimize** this criterion to ensure affordability during the exchange.
- **Quality of Life (Life Expectancy)**: This criterion evaluates the overall quality of life in the host city or country, measured by life expectancy. The scale ranges from 60 to 90 years, where higher values indicate a better quality of life. The goal is to **maximize** this criterion, as the student may want to live in the area after the exchange and seeks a healthy, stable environment.
- **Average Salary after Graduation**: This criterion measures the average salary that graduates from the university can expect to earn in their field, typically after completing a master's program. The scale ranges from 0 to 100,000€, with higher salaries being more desirable. The goal is to **maximize** this criterion to ensure the student has strong future financial prospects.
- **Distance from Barcelona**: This variable measures the distance between Barcelona and the university, with the scale ranging from 0 to 1,500 kilometers. A larger distance indicates a more unique and immersive experience. The goal is to **maximize** this criterion, as the student is looking for a more distinct cultural experience abroad.

In a multicriteria decision making method, three types of thresholds can be used to express the decision maker's preferences and tolerance for imprecision:

Criterion	Range	Goal
Reputation of the University	1 to 500	**Minimize**
Scholarships and Financial Support	0 to 15,000€	**Maximize**
Monthly Cost of Living	500€ to 2,500€	**Minimize**
Quality of Life (Life Expectancy)	60 to 90 years	**Maximize**
Average Salary after Graduation	0 to 100,000€	**Maximize**
Distance from Barcelona	0 to 1,500 km	**Maximize**

Table 1: Criteria summary Table

- **Indifference threshold (q)**: The largest difference in performance between two alternatives for which they are still considered essentially equivalent for a given criterion.
- **Preference threshold (p)**: The smallest difference for which one alternative is clearly preferred over the other on a specific criterion.
- **Veto threshold (v)**: A critical difference above which a poor performance on a criterion is sufficient to veto the outranking of one alternative over another, regardless of other favorable criteria.

Table 2 shows the values of the different thresholds as a summary, and the explanations are the following:

1. **Reputation of the University (Minimize)**:
   $q = 5$, $p = 30$, $v = 100$
   Small differences in university ranking (less than 5) are negligible; preferences appear above 30 ranks, while a gap of 100 ranks can act as a veto.
2. **Scholarships and Financial Support (Maximize)**:
   $q = 500$, $p = 1,500$, $v = 7,000$
   Minor differences under 500€ are treated as indifference; a difference of 1,500€ shows preference, while a shortfall of more than 7,000€ may trigger a veto.
3. **Monthly Cost of Living (Minimize)**:
   $q = 100$, $p = 500$, $v = 1,000$
   Costs differing by less than 100€ are negligible; differences over 500€ indicate preference, and a cost exceeding another by 1,000€ can be vetoed.
4. **Quality of Life (Life Expectancy) (Maximize)**:
   $q = 1$ year, $p = 5$ years, $v = 10$ years
   Life expectancy differences under 1 year are insignificant; a gap of 5 years shows preference, while a deficit of 10 years may result in a veto.
5. **Average Salary after Graduation (Maximize)**:
   $q = 2,000$, $p = 10,000$, $v = 25,000$
   Differences below 2,000€ are negligible; a salary 10,000€ higher is clearly preferred, and a 25,000€ difference may lead to a veto.
6. **Distance from Barcelona (Maximize)**:
   $q = 50$ km, $p = 300$ km, $v = 700$ km
   Small changes within 50 km are indifferent; preference begins at 300 km, and a 700 km deficit can veto an outranking.

Criterion	Indifference (q)	Preference (p)	Veto (v)
Reputation of the University	5	30	100
Scholarships and Financial Support	500€	1,500€	7,000€
Monthly Cost of Living	100€	500€	1,000€
Quality of Life (Life Expectancy)	1 year	5 years	10 years
Average Salary after Graduation	2,000€	10,000€	25,000€
Distance from Barcelona	50 km	300 km	700 km

Table 2: Threshold values (indifference, preference, veto) for each criterion

1.3 Definition of alternatives to the problem

University	World Ranking (0–500)	Scholarships (€)	Cost of Living (€)	Life Expectancy (Years)	Salary After Graduation (€)	Distance from Barcelona (km)
Albert-Ludwigs-Universität Freiburg	192	10,000	1,200	81	55,000	1,100
University of Buenos Aires	71	0	600	76	18,000	10,000
Université Catholique de Louvain	170	5,000	1,000	80	42,000	1,200
Politecnico di Milano	123	6,000	1,200	82	32,000	1,200
Georgia Institute of Technology	88	15,000	1,500	79	85,000	8,000
Technische Universität München	37	12,000	1,300	80	60,000	1,400
Pontificia Universidad Católica de Chile	121	8,000	800	81	28,000	10,000
ETH Zurich	7	10,000	1,800	84	90,000	1,000
Aalto University	109	5,000	1,100	82	48,000	2,000
National Taiwan University of Science and Technology	387	4,000	700	80	30,000	11,000
University of Edinburgh	22	7,000	1,300	82	50,000	1,500
University of Oulu	313	6,000	900	81	45,000	2,100

Table 3: Universities Compared by Criteria

1.4 Type of Problematic to Solve

In this case, the decision-making problem corresponds to a **ranking problem**. The student (DM) is trying to decide which exchange universities are the most appealing based on several personal priorities, as described in the criteria section.

It is not a **sorting problem**, since alternatives are not classified into 'good' or 'bad' categories as all experiences are unique and potentially enriching in different ways. It is not a **choice problem**, as the goal is not to select a single best alternative, but rather to obtain an outranking relation list of universities that align with the student's preferences. This list will serve as a starting point to explore the most suitable options more closely.

Thus, this situation is best modeled as a **ranking problem**, where multiple alternatives are evaluated across several criteria to produce a ranking reflecting the student's individual priorities.

1.5 MCDA Method Proposal

To address the decision-making problem described, it is necessary to select an appropriate Multi-Criteria Decision Analysis (MCDA) method. For this purpose, we propose using the **MCDA-MSS** (Multi-Criteria Decision Aid Method Selection Software), a tool designed to recommend suitable MCDA methods based on the specific characteristics of the decision problem. The tool guides users through a series of structured questions, helping to identify the most appropriate method to apply. Here we will describe the questions and answers that will be answered to obtain our method.

- **Problem statement:** The decision recommendation type is a **ranking** problem. The decision-maker (a student) is not looking to select a single best alternative or to classify options into categories but rather to establish a prioritized list of preferred exchange universities. Since no option is inherently bad, and the goal is to explore a range of promising alternatives, ranking best reflects the objective of supporting a nuanced and comparative evaluation.

- **Order of alternatives:** A **partial** order of alternatives is requested. Rather than forcing a strict ranking among all universities (which may not be meaningful given close or incomparable performances), a partial order allows for some alternatives to be considered equally preferable or incomparable.
- **Scale leading the recommendation:** A **ordinal** scale is used. The goal is to prioritize alternatives rather than to produce a numerical utility score. Since the student is more interested in relative preferences than precise quantitative rankings, an ordinal interpretation of results is appropriate.
- **Set of alternatives:** The set of alternatives is considered **stable**. The student is analyzing a known, fixed list of exchange universities and does not expect new alternatives to appear during the decision process.
- **Criteria structure:** The criteria structure is **flat**. All evaluation criteria (e.g., cost of living, scholarships, salary after graduation) are considered at the same level without being grouped or organized into a hierarchy.
- **Evaluation of alternatives on the criteria:** The performances are assumed to be **deterministic**. The student uses available data and averages to compare the universities, without incorporating uncertainty or probabilistic models.
- **Criteria preference direction:** The preference direction for each criterion is **known**. For example, higher scholarships, lower cost of living, and greater post-graduation salaries are clearly preferred.
- **Criteria set completeness:** The set of criteria is **complete**. The student has considered all relevant aspects that influence their decision and does not expect to add more criteria.

With this results, the obtained method is **ELECTRE III for Ranking**.

2 Part 2: Implementation of the Decision Problem

The implementation of the decision problem will be carried out using Python, specifically utilizing the `mcda` libraries introduced in the subject. The decision-making model is based on the ELECTRE III method, as identified in Part 1. This process involves translating the theoretical components of the decision problem, such as criteria, thresholds, and alternatives into a computational framework using these tools in order to support the decision maker, in this case, the student.

2.1 Data Preparation and Model Parameters

The alternatives (universities) are represented in a matrix, where each row corresponds to a university and each column to one of the six evaluation criteria: reputation (world ranking), scholarships, cost of living, life expectancy, salary after graduation, and distance from Barcelona. These criteria were treated as independent.

Threshold parameters, preference thresholds (p), indifference thresholds (q), and veto thresholds (v) were defined for each criterion to model the decision maker's sensitivity to differences in performance across alternatives. These threshold values are summarized in Table 2.

Two experiments were conducted using different sets of weights and thresholds to reflect varying decision-maker preferences. Table 4 summarizes the weight values used in each scenario.

Criterion	Initial Weights	Life After Graduation Focus
World Ranking	0.25	0.10
Scholarships (€)	0.15	0.10
Cost of Living (€)	0.15	0.20
Life Expectancy (Years)	0.10	0.20
Salary After Graduation (€)	0.25	0.30
Distance from Barcelona (km)	0.10	0.10

Table 4: Criteria Weights for the Two Experiments

These experiments will allow the model to explore how shifting priorities, such as valuing quality of life and financial outcomes more highly can influence the final ranking of alternatives.

2.2 ELECTRE III Methodology Implementation

The ELECTRE III method is a multi-criteria decision-making (MCDM) technique based on pairwise comparisons of alternatives, incorporating decision-maker preferences and thresholds to handle imperfect knowledge. The implementation of this methodology consists of the following steps, which translate the theoretical framework into a structured computational process:

1. **Construction of the Performance Table**

   The first step involves building the performance table that contains all relevant data for evaluation. Alternatives (universities) are represented as rows, and evaluation criteria as columns. Each cell in the table corresponds to the performance of a university with respect to a specific criterion. The criteria, their weights, the scale and the preference directions (maximization or minimization) are defined based on the problem context. These are drawn from Tables 1, 3, and 4.

2. **Computation of Concordance, Discordance, and Adjacency Matrices**

   Using the thresholds provided in Table 2, three key matrices are generated:

   - **Concordance Matrix:** Measures the degree to which one alternative is at least as good as another across all criteria, taking into account the weights and indifference/preference thresholds.
   - **Discordance Matrix:** Identifies whether there is a strong opposition (discordance) to the assertion that one alternative is better than another on specific criteria.
   - **Adjacency Matrix:** A binary representation used to capture the dominance relationships derived from the concordance and discordance relations.

   These matrices help formalize the outranking relationships of the ELECTRE III method.

3. **Construction of the Credibility Matrix**

   The credibility matrix synthesizes the information from the concordance and discordance matrices to determine the overall strength of the outranking relation between pairs of alternatives. It is computed using a non-compensatory aggregation rule that incorporates the veto thresholds to penalize significant disadvantages in specific criteria. In this step multiple threshold configurations may be tested here to reflect different sensitivity assumptions.

4. **Distillation Procedures (Ascending and Descending)**

   Once the credibility matrix is established, ELECTRE III applies two distillation procedures:

   - **Descending distillation:** Identifies the best alternatives by eliminating the weakest ones in successive iterations.
   - **Ascending distillation:** Identifies the worst alternatives by successively removing the top-ranked ones.

   These procedures generate two partial preorders which are then merged to produce a final ranking of alternatives.

	World Ranking	Scholarships (€)	Cost of Living (€)	Life Expectancy (Years)	Salary After Graduation (€)	Distance from Barcelona (km)
Albert-Ludwigs-Universität Freiburg	192	10000	1200	81	55000	1100
University of Buenos Aires	71	0	600	76	18000	10000
Université Catholique de Louvain	170	5000	1000	80	42000	1200
Politecnico di Milano	123	6000	1200	82	32000	1200
Georgia Institute of Technology	88	15000	1500	79	85000	8000
Technische Universität München	37	12000	1300	80	60000	1400
Pontificia Universidad Católica de Chile	121	8000	800	81	28000	10000
ETH Zurich	7	10000	1800	84	90000	1000
Aalto University	109	5000	1100	82	48000	2000
National Taiwan University of Science and Technology	387	4000	700	80	30000	11000
University of Edinburgh	22	7000	1300	82	50000	1500
University of Oulu	313	6000	900	81	45000	2100

Figure 1: Performance Table

	Albert-Ludwigs-Universität Freiburg	University of Buenos Aires	Université Catholique de Louvain	Politecnico di Milano	Georgia Institute of Technology	Technische Universität München	Pontificia Universidad Católica de Chile	ETH Zurich	Aalto University	National Taiwan University of Science and Technology	University of Edinburgh	University of Oulu
Albert-Ludwigs-Universität Freiburg	1.00000	0.5000	0.7725	0.7300	0.250	0.40625	0.5375	0.45000	0.65000	0.7500	0.65000	0.82500
University of Buenos Aires	0.50000	1.0000	0.5250	0.5000	0.550	0.27500	0.5000	0.25000	0.50000	0.4250	0.25000	0.50000
Université Catholique de Louvain	0.60000	0.5375	1.0000	0.6500	0.250	0.29000	0.4625	0.27500	0.50000	0.8250	0.28750	0.79375
Politecnico di Milano	0.60000	0.5000	0.7125	1.0000	0.250	0.29000	0.6375	0.32500	0.56000	0.7500	0.32500	0.57500
Georgia Institute of Technology	0.90000	0.6300	0.8500	0.8750	1.000	0.71250	0.7250	0.55625	0.83750	0.7500	0.66250	0.82500
Technische Universität München	1.00000	0.7500	0.9250	0.9750	0.500	1.00000	0.7500	0.42500	0.83750	0.7500	0.85500	0.78750
Pontificia Universidad Católica de Chile	0.60000	0.7125	0.7500	0.9375	0.350	0.35000	1.0000	0.30000	0.68000	0.9000	0.50000	0.75000
ETH Zurich	0.83000	0.7500	0.7900	0.7900	0.675	0.60000	0.7500	1.00000	0.75000	0.7500	0.75000	0.75000
Aalto University	0.69375	0.5000	1.0000	0.9250	0.340	0.35000	0.6750	0.32500	1.00000	0.7875	0.60000	0.86750
National Taiwan University of Science and Technology	0.35000	0.7500	0.4250	0.5750	0.350	0.35000	0.6000	0.27500	0.40000	1.0000	0.32500	0.35000
University of Edinburgh	0.75625	0.7500	0.9250	1.0000	0.500	0.60000	0.6750	0.47500	0.86250	0.7500	1.00000	0.78750
University of Oulu	0.35000	0.5750	0.7500	0.7500	0.250	0.35000	0.5000	0.30000	0.71875	0.8625	0.58125	1.00000

Figure 2: Concordance Matrix on Experiment 1

5. **Exploitation and Visualization**

   The final step involves synthesizing the results into an interpretable format. The exploitation plot visually represents the position of each alternative in both ascending and descending orders, highlighting rank consistency and ambiguity. This visualization aids the decision maker in interpreting the results and making informed choices.

2.3 Implementation of ELECTRE III Decision Making Problem

2.3.1 Experiment 1: Initial Weight Setup

This first experiment uses a balanced set of weights that reflect a generalist perspective, where each criterion is given moderate importance. The aim is to establish a baseline ranking of alternatives by assuming that the decision maker values all aspects of the exchange experience relatively equally.

As shown in the Performance Table in Figure 1, the values of each criteria on each alternative match those defined in Table 1.

From the performance table we can not extract any relevant conclusions yet. However, after computing the concordance matrix we can start seeing some relevant insights on which alternative may be better for the decision maker.

For instance, if we look row wise, we can observe some good candidates that show that at most pairwise comparisons, they are at least as good as the other alternatives. Among these, we can see that TU München in all comparisons except one exceed 0.5. Similar behaviours can be seen on ETH Zurich, University of Edinburgh and GeorgiaTech.

If we plot the results of the concordance matrix using a threshold of 0.75 we can observe the previous insights that were seen. As, in this case we can see that TU München, ETH Zurich and Georgiatech outrank most of the other alternatives.

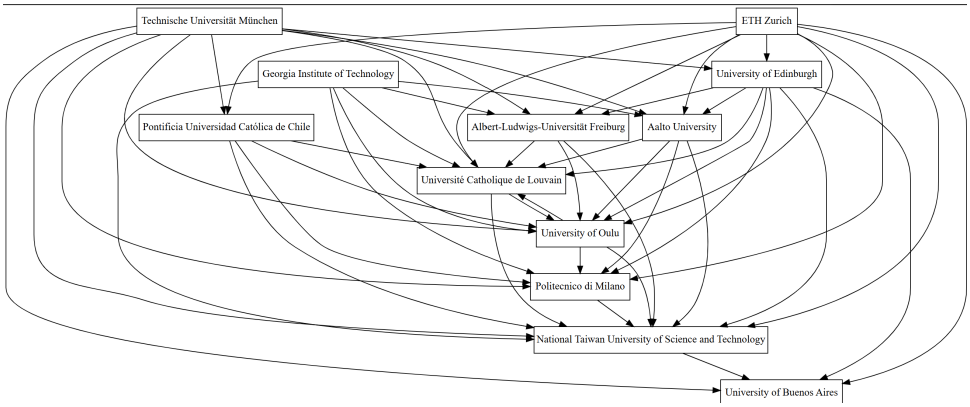


Figure 3: Adjacency Matrix built on concordance matrix with a threshold of 0.75

	Albert-Ludwigs-Universität Freiburg	University of Buenos Aires	Université Catholique de Louvain	Politecnico di Milano	Georgia Institute of Technology	Technische Universität München	Pontificia Universidad Católica de Chile	ETH Zurich	Aalto University	National Taiwan University of Science and Technology	University of Edinburgh	University of Oulu
Albert-Ludwigs-Universität Freiburg	1.000	0.000	0.772	0.730	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
University of Buenos Aires	0.000	1.000	0.056	0.182	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Université Catholique de Louvain	0.545	0.000	1.000	0.650	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Politecnico di Milano	0.200	0.000	0.712	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Georgia Institute of Technology	0.900	0.000	0.850	0.875	1.000	0.712	0.000	0.340	0.838	0.000	0.662	0.825
Technische Universität München	1.000	0.000	0.925	0.975	0.000	1.000	0.000	0.000	0.837	0.000	0.855	0.000
Pontificia Universidad Católica de Chile	0.000	0.712	0.750	0.938	0.000	0.000	1.000	0.000	0.680	0.000	0.006	0.750
ETH Zurich	0.830	0.000	0.790	0.790	0.000	0.600	0.000	1.000	0.000	0.000	0.750	0.000
Aalto University	0.694	0.000	1.000	0.925	0.000	0.000	0.000	0.000	1.000	0.000	0.279	0.867
National Taiwan University of Science and Technology	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000	0.000	0.200
University of Edinburgh	0.756	0.000	0.925	1.000	0.000	0.545	0.000	0.000	0.862	0.000	1.000	0.787
University of Oulu	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000

Figure 4: Credibility Matrix

After obtaining the concordance matrix, the next step is to compute the discordance matrix and the credibility matrix. In this case, we computed both but just showed the results of the credibility matrix as it is the aggregation of both the concordance and discordance matrices.

As shown in Figure 4, the credibility matrix reveals several key insights. Technische Universität München, ETH Zurich, and Georgia Institute of Technology consistently dominate other universities, indicating strong overall performance. For example, TUM shows high credibility over Politecnico di Milano (0.975) and Université Catholique de Louvain (0.925).

Université Catholique de Louvain and Aalto University also perform well, each showing dominance over several peers. In contrast, universities like University of Buenos Aires, National Taiwan University of Science and Technology, and University of Oulu have low credibility values, suggesting weaker performance in the evaluated criteria.

If we plot the results with a threshold of 0.6, as seen in Figure 5, we can see the results explained previously, with more dominance on ETH Zurich and Georgiatech over TU München.

Finally, the distillation process will help us get a clearer picture of how the alternatives rank by looking at them from two angles ascending and descending.

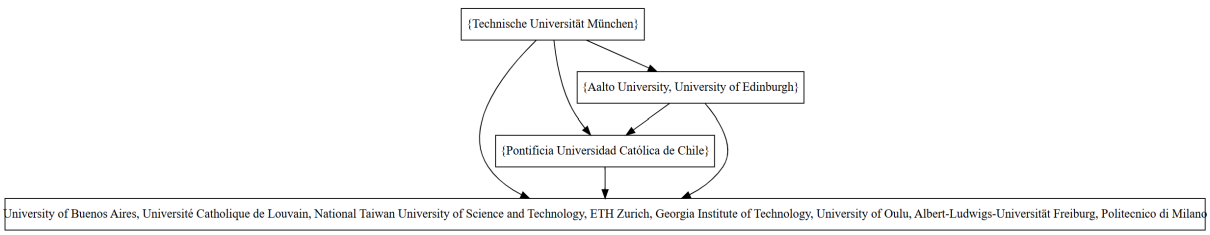


Figure 6: Descending Distillation Results

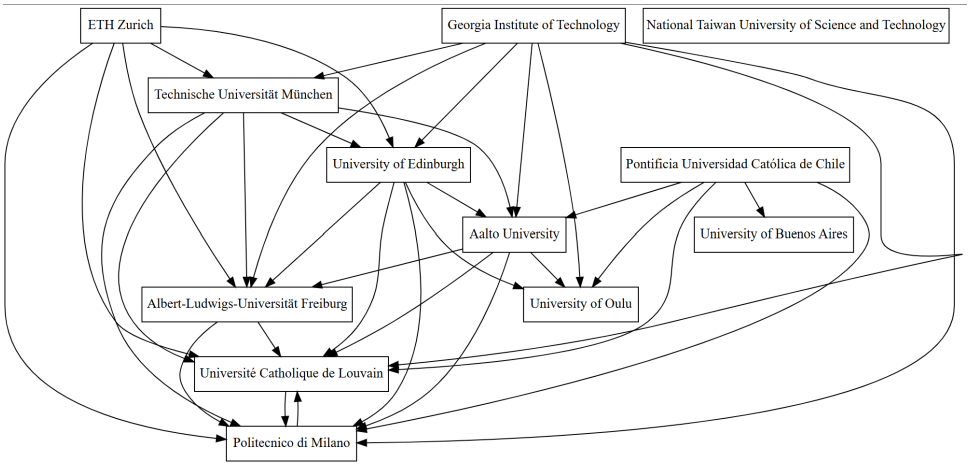


Figure 5: Adjacency Matrix Plot from Credibility matrix with threshold of 0.6

The **ascending distillation** (Figure 7) starts by grouping the less preferred universities. The first group is quite large and includes schools like the University of Buenos Aires, Georgia Institute of Technology, and ETH Zurich. Then, as we move up the ranking, we find smaller groups with universities like Albert-Ludwigs-Universität Freiburg, Université Catholique de Louvain, and Politecnico di Milano, which seem to be better positioned overall.

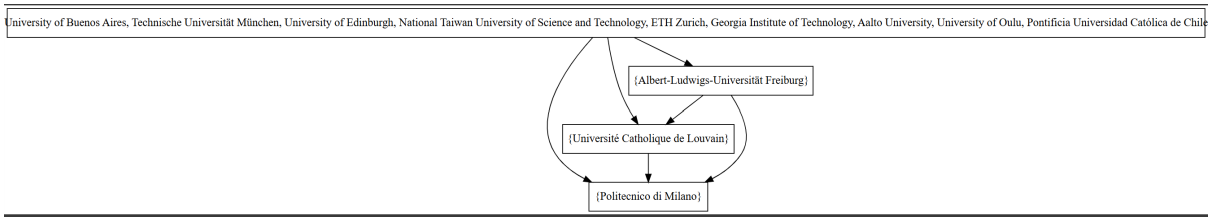


Figure 7: Ascending Distillation Results

On the other hand, the **descending distillation** (Figure 6) starts by picking out the top performers. Here, Technische Universität München stands out alone at the very top. Right after, Aalto University and the University of Edinburgh form the next best group, followed by Pontificia Universidad Católica de Chile in its own group. The rest of the universities fall into a larger group towards the bottom of the ranking.

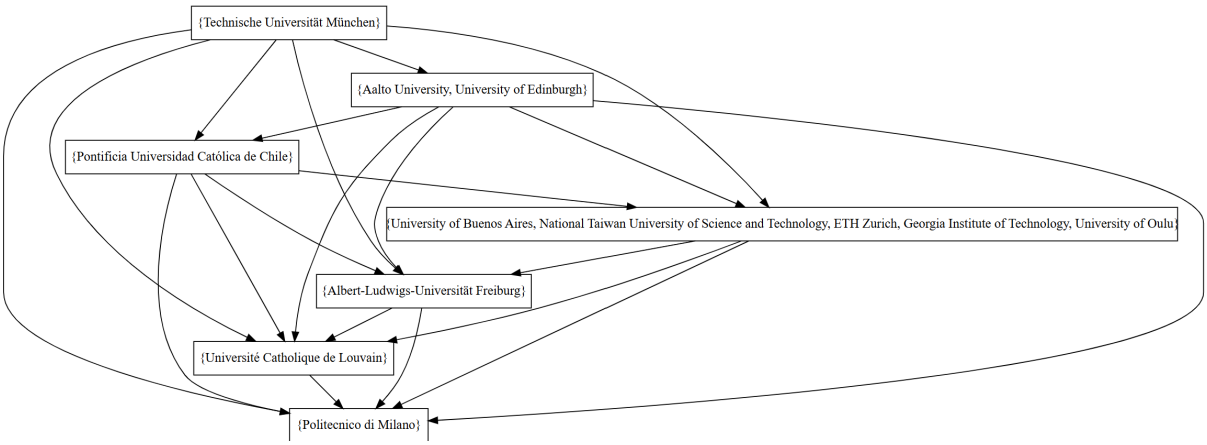


Figure 8: Exploitation Results for Experiment 1

Looking at both distillations and at the final exploit ranking in the Figure 8 together gives us a fuller

story: Technische Universität München clearly leads the alternatives, while a bigger cluster of universities share the lower rankings. Surprisingly, ETH Zurich and GeorgiaTech are placed significantly lower here.

2.3.2 Experiment 2: Initial Weight Setup

In the second experiment, both the weight distribution and the threshold values are adjusted to reflect a decision maker who prioritizes long-term benefits after graduation. More importance is assigned to financial outcomes and quality of life indicators such as salary, cost of living, and life expectancy. This setup allows the model to analyze how changes in both the criteria weights and sensitivity thresholds influence the final ranking of the alternatives (see Table 4 for weights and Table 5 for thresholds).

Criterion	Indifference (q)	Preference (p)	Veto (v)
Reputation of the University	10	25	90
Scholarships and Financial Support	700€	1,200€	6,000€
Monthly Cost of Living	50€	300€	700€
Quality of Life (Life Expectancy)	0.5 year	3 years	8 years
Average Salary after Graduation	1,000€	7,000€	20,000€
Distance from Barcelona	75 km	250 km	600 km

Table 5: Adjusted threshold values for each criterion in Experiment 2

In this second experiment, the threshold values have been adjusted to reflect a decision maker focused on long-term post-graduation benefits. Financial factors such as *scholarships*, *cost of living*, and *average salary* have lower thresholds in order to indicate a greater sensitivity to economic conditions that affect future stability. The thresholds for *quality of life* as life expectancy were also reduced which empathize its role in sustained well-being. Also, the *university reputation* and *distance from Barcelona* thresholds have been changed to make a more nuanced evaluation of academic prestige and a greater willingness to consider distant options when justified by long-term gains. Table 5 shows the new values.

The procedure to evaluate the new situation is the same as that of the first experiment, were the initial concordance and discordance are computed, then the credibility matrix and finally the distillation results and the exploitation results.

Figure 9 shows the adjacency matrix plot obtained from the concordance matrix with a cut threshold of 0.75. From this figure we can see that the results are highly similar to those on the first experiment where the TU München university, ETH Zurich and GeorgiaTech outrank most of the other alternatives.

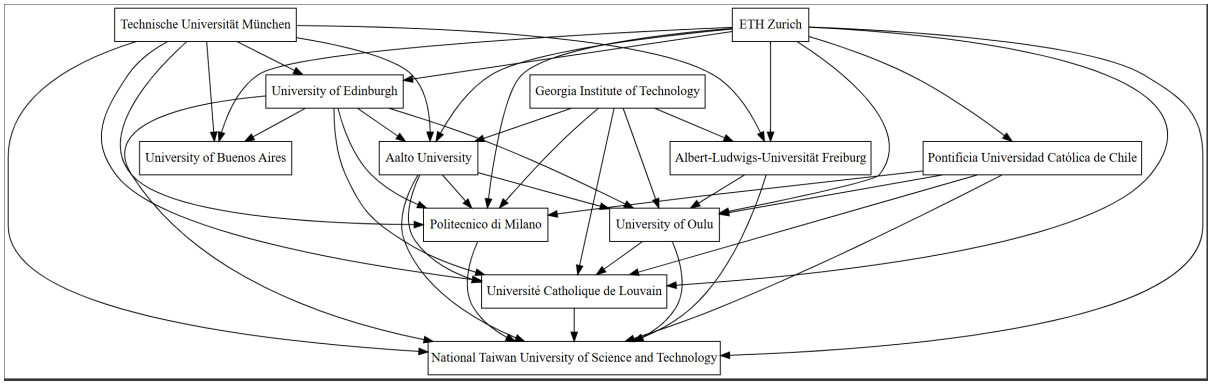


Figure 9: Concordance Results for Experiment 2 with a cut threshold of 0.6

To deepen the analysis, a credibility matrix was computed too, but similarly to the results on the first experiment, we obtain similar conclusions from those obtained in the concordance plot. As shown in figure 10 we can observe similar results to those in Figure 5 in the first experiment.

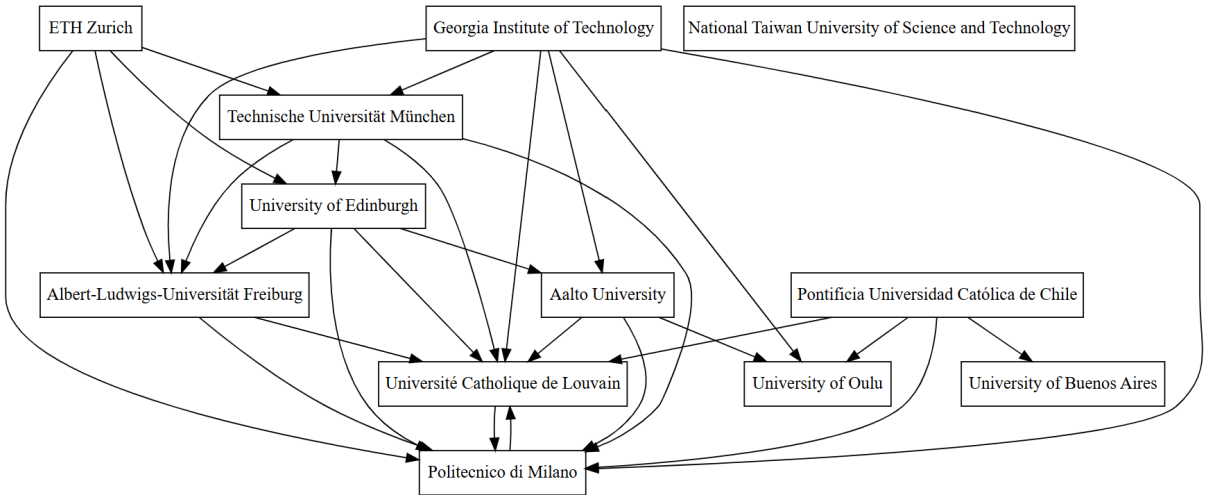


Figure 10: Credibility Results for Experiment 2

The final results of the ELECTRE III method for Experiment 2 offer a clear narrative through the ascending and descending distillation procedures. As shown in the ascending distillation (Figure 12), a leading group of universities—including the University of Buenos Aires, Georgia Institute of Technology, and Technische Universität München, quickly rise to the top, which indicate a strong and consistent alignment with the decision maker's long-term goals. On the other hand, universities like Politecnico di Milano appear only in the later stages of the process, suggesting they fall short in areas prioritized in this experiment.

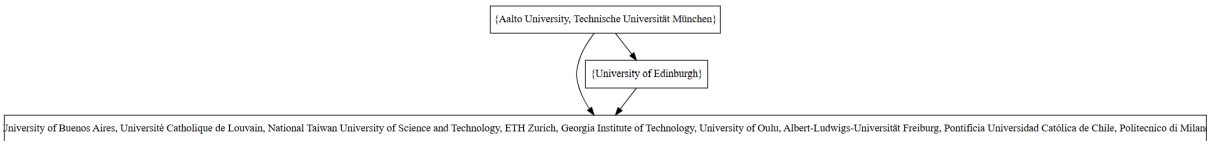


Figure 11: Descending Distillation Results for Experiment 2

The descending distillation (Figure 11) presents a complementary perspective. Here, Technische Universität München and Aalto University are ranked at the top, reinforcing their strong standing. Meanwhile, institutions like Politecnico di Milano and Université Catholique de Louvain are grouped towards the lower end, consistent with their placement in the ascending distillation.

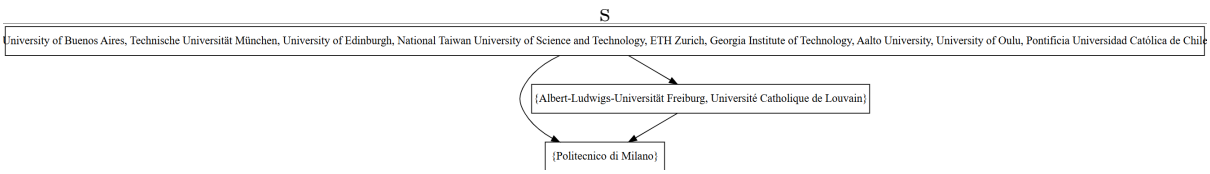


Figure 12: Ascending Distillation Results for Experiment 2

Looking at the full picture, as seen in Figure 13, the two universities that clearly stand out are Technische Universität München and Aalto University. Both consistently emerge at the top across both the distillation procedures, showing strong performance on the criteria that matter most in this experiment, particularly long-term financial outcomes and quality of life.

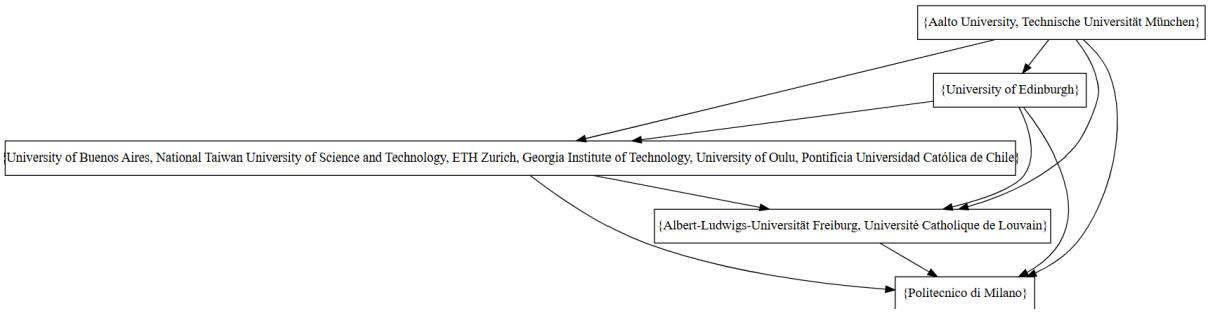


Figure 13: Exploitation Results for Experiment 2

2.4 Discussion of results & Recommendation to the decision maker

Looking at the results, some universities consistently come up as strong options in both experiments. The ETH Zurich, Georgia Institute of Technology, and Technical University of Munich stand out as top choices regardless of how the criteria are weighted. Other universities like Politecnico di Milano, Albert-Ludwigs-Universität Freiburg, Université Catholique de Louvain, and the University of Oulu also perform well but don't quite reach the same level as the top three.

In the first experiment, where all criteria were treated fairly equally, the outcome was quite clear. After the distillation process, the Technical University of Munich clearly outranked the other alternatives, making it the best pick under those conditions. But in the second experiment, where more importance was given to quality of life and long-term financial benefits, both the Technical University of Munich and Aalto University in Finland emerged as the leaders. This makes sense because Aalto University offers a great quality of life and good salary prospects after graduation, which became more important with the changed priorities.

These differences show how much the decision maker's preferences can affect the final ranking. So, it's really important to think carefully about what matters most in the decision. If the goal is a balanced approach, the Technical University of Munich is the clear winner. But if quality of life and financial outcomes are the priority, both the Technical University of Munich and Aalto University deserve consideration.

3 Conclusions & Personal evaluation

This project has demonstrated the effectiveness of the ELECTRE III method as a decision support tool for complex, multicriteria problems such as selecting a university for an exchange program. Through the careful formulation of the decision problem, the definition of criteria, and the application of thresholds for indifference, preference, and veto, we were able to provide a structured and insightful ranking of alternatives.

The final results highlight the importance of considering multiple perspectives when making decisions of this nature. While universities like ETH Zurich and Georgia Institute of Technology are globally prestigious, the analysis revealed that factors such as cost of living and distance can significantly influence the final ranking which revealed that probably the best alternatives for the decision maker were the TU München and Aalto University, depending on the experiment.

From a personal perspective, this project has been an enriching learning experience. It allowed us to deepen our understanding of multicriteria decision analysis and the practical application of the ELECTRE III method using Python. We gained hands-on experience in translating a theoretical framework into a working model, handling real-world data, and interpreting the results in a meaningful way.

One of the most valuable lessons we learned is that decision-making is rarely about finding a perfect solution. Rather, it is about understanding trade-offs, aligning choices with specific priorities, and ensuring that decisions reflect the decision maker's values and preferences. This insight is highly relevant beyond the scope of this project and can be applied in other domains, from business strategy to personal

life decisions.

In conclusion, this project not only provided a rigorous ranking of universities for the exchange program but also served as a valuable exercise in applying advanced decision-making techniques. It has strengthened our ability to tackle complex problems using a structured analytical approach, skills that we believe will be highly beneficial in our future academic and professional journeys.